

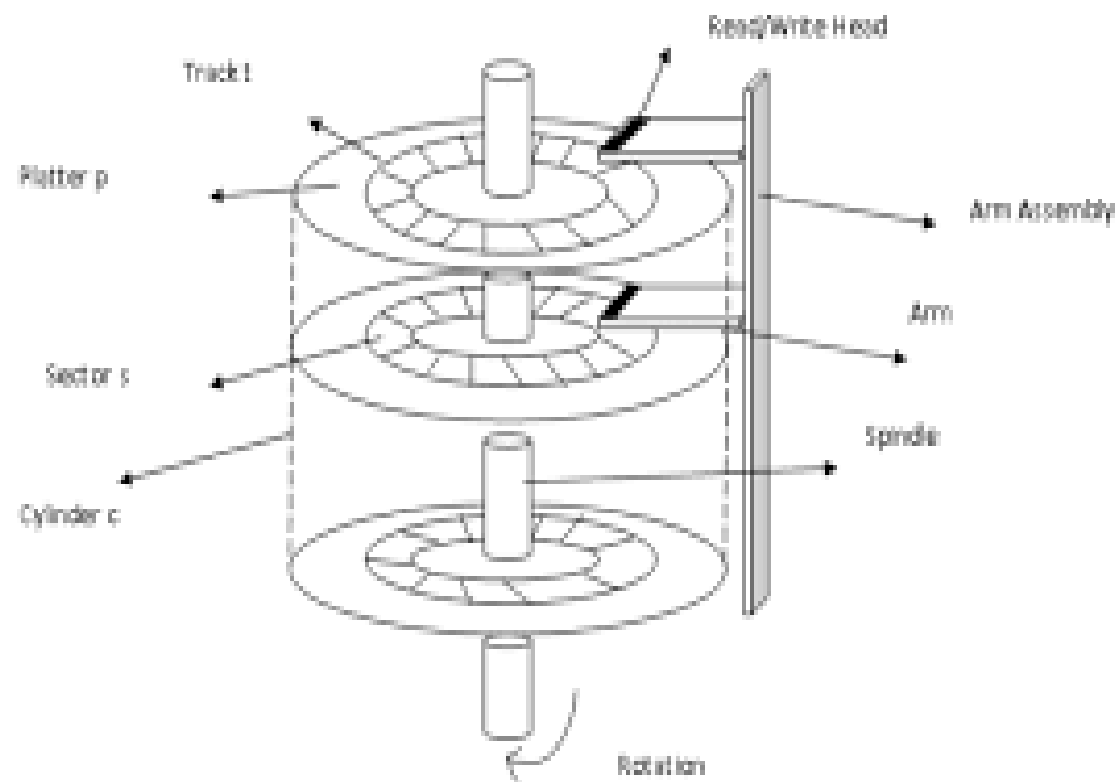
DISK SCHEDULING ALGORITHMS

INTRODUCTION

- The requests made by a process may be located at different sectors on different tracks.
- Due to this , the seek time might increase. These algorithms help in minimizing the seek time by ordering the requests made by the processes.
- Disk scheduling algorithms manage how data is read from and written to a computer's hard disk.
- These algorithms help determine the order in which disk read and write requests are processed.
- Disk scheduling is also known as I/O Scheduling.
- The main goals of disk scheduling are to optimize the performance of disk operations, reduce the time it takes to access data and improve overall system efficiency.

IMPORTANT TERMS RELATED TO DISK SCHEDULING ALGORITHMS

- **Seek Time** - It is the time taken by the disk arm to locate the desired track.
- **Rotational Latency** - The time taken by a desired sector of the disk to rotate itself to the position where it can access the Read/Write heads is called Rotational Latency.
- **Transfer Time** - It is the time taken to transfer the data requested by the processes.
- **Disk Access Time** - Disk Access time is the sum of the Seek Time, Rotational Latency, and Transfer Time.



Disk Scheduling Algorithms

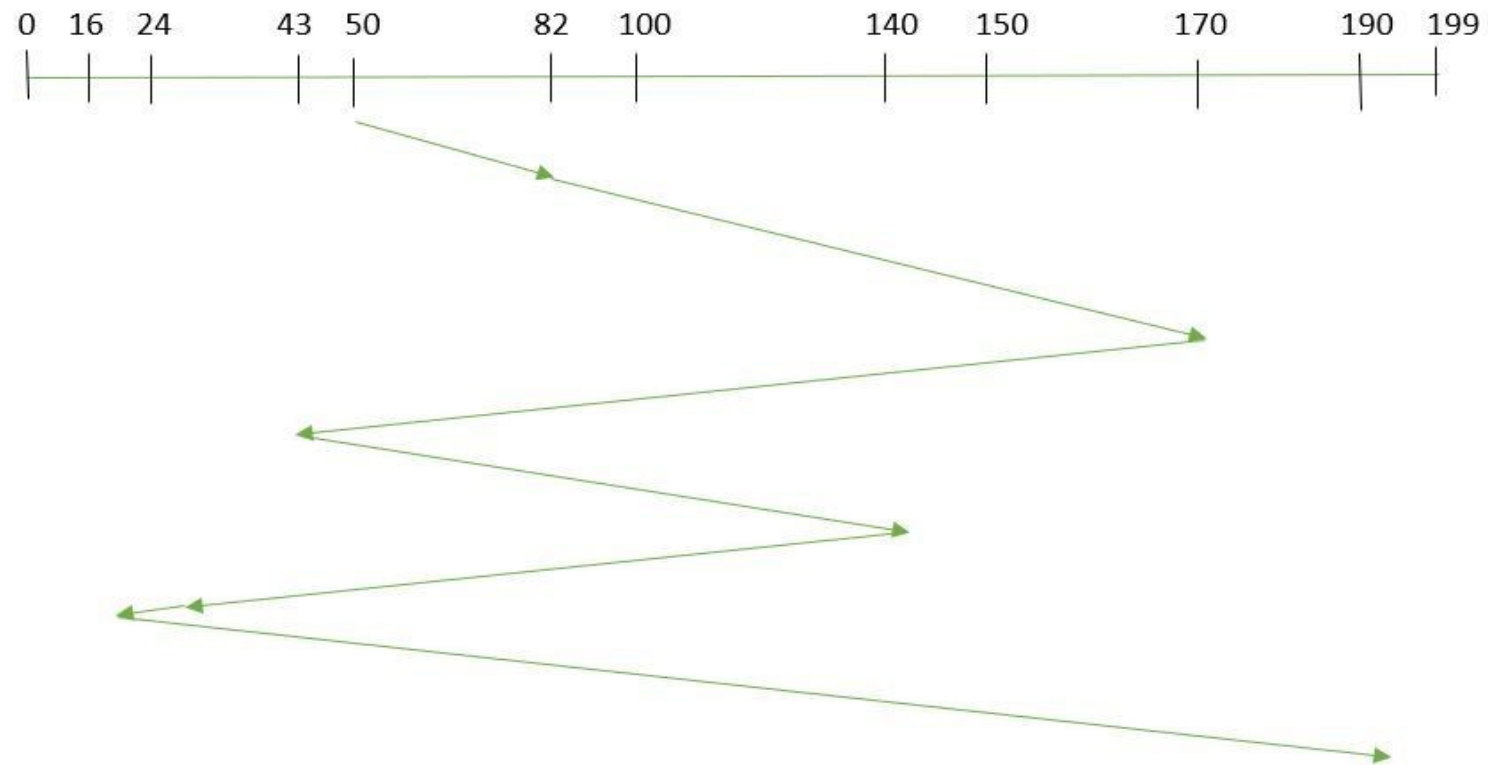


Disk Scheduling Algorithms

There are several Disk Scheduling Algorithms :

- FCFS (First Come First Serve)
- SSTF (Shortest Seek Time First)
- SCAN
- C-SCAN
- LOOK
- C-LOOK

1. FCFS



Suppose the order of request is- (82,170,43,140,24,16,190) and current position of Read/Write head is: 50

So, total overhead movement (total distance covered by the disk arm)
 $= (82-50)+(170-82)+(170-43)+(140-43)+(140-24)+(24-16)+(190-16)$
 $= 642$

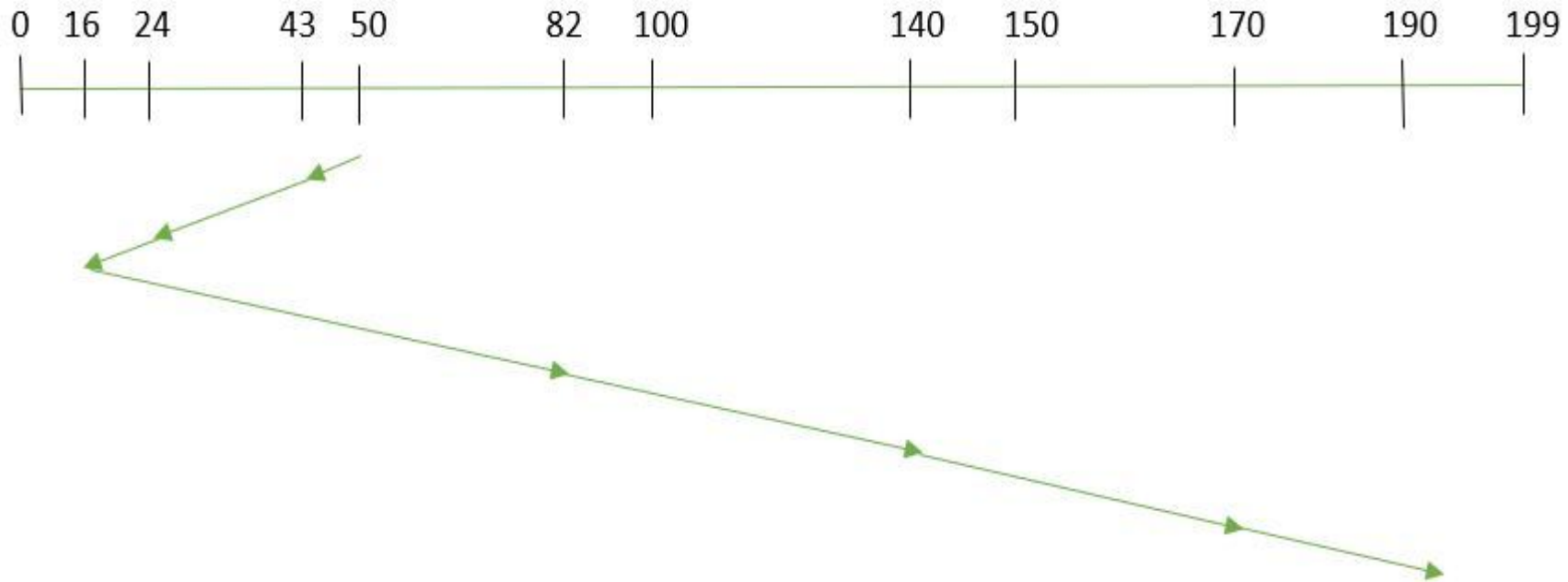
Advantages of FCFS

- Every request gets a fair chance
- No indefinite postponement

Disadvantages of FCFS

- Does not try to optimize seek time
- May not provide the best possible service

2. SSTF (Shortest Seek Time First)



Suppose the order of request is- (82,170,43,140,24,16,190) and current position of Read/Write head is: 50

$$\begin{aligned} &\text{total overhead movement (total distance covered by the disk arm)} \\ &= (50-43) + (43-24) + (24-16) + (82-16) + (140-82) + (170-140) + (190-170) \\ &= 208 \end{aligned}$$

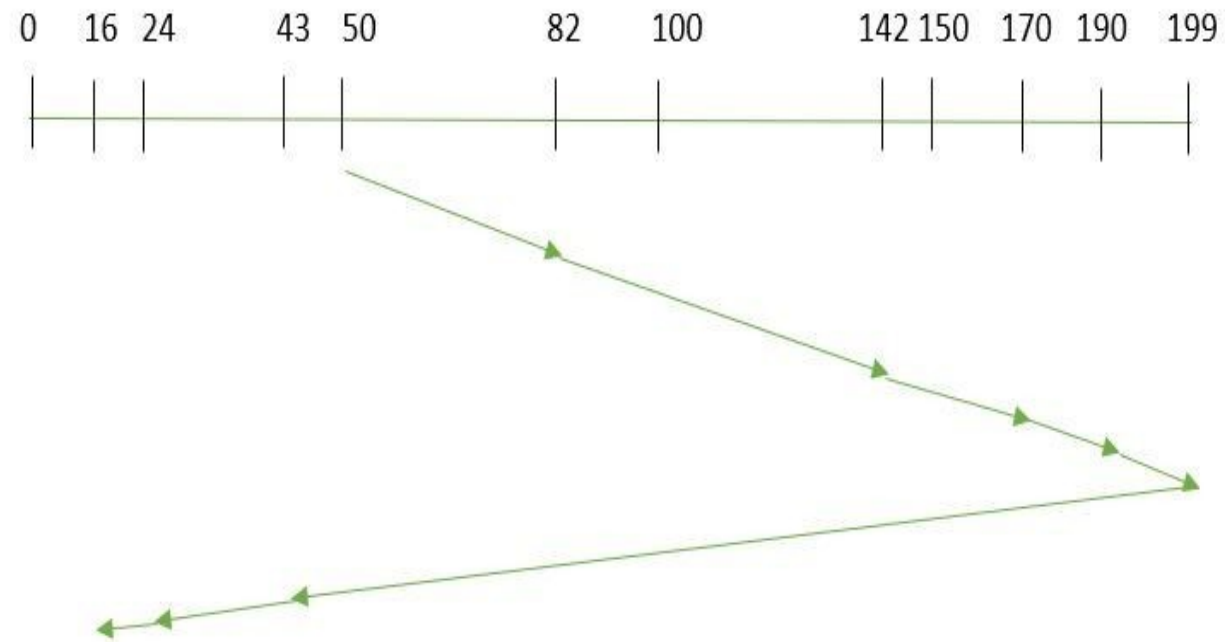
Advantages of Shortest Seek Time First

- The average Response Time decreases
- Throughput increases

Disadvantages of Shortest Seek Time First

- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has a higher seek time as compared to incoming requests
- The high variance of response time as SSTF favors only some requests

3. SCAN



Suppose the requests to be addressed are-82,170,43,140,24,16,190 and the Read/Write arm is at 50, and it is also given that the disk arm should move "**towards the larger value**".

Therefore, the total overhead movement (total distance covered by the disk arm) is calculated as

$$= (199-50) + (199-16) = 332$$

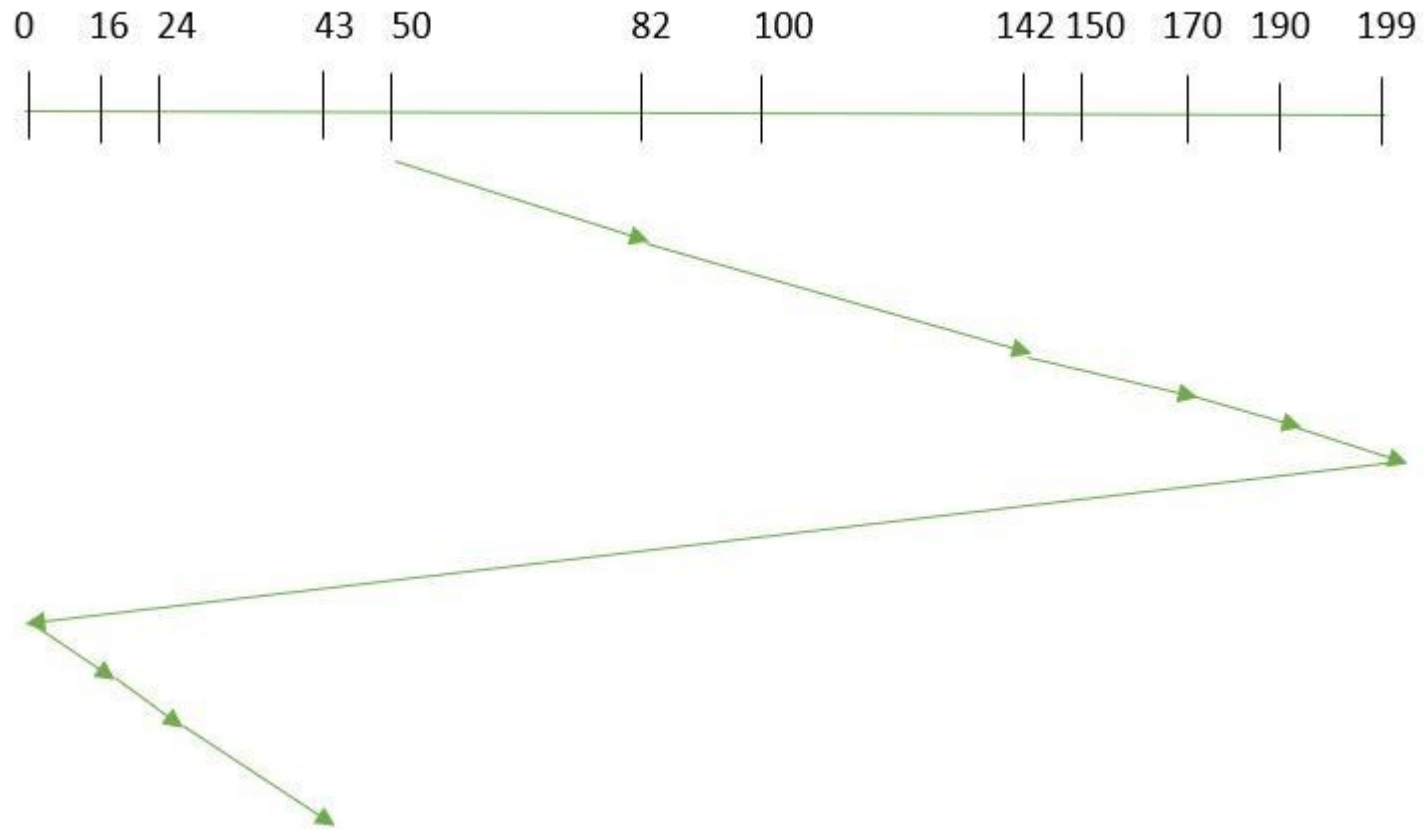
Advantages of SCAN Algorithm

- High throughput
- Low variance of response time
- Average response time

Disadvantages of SCAN Algorithm:

- Long waiting time for requests for locations just visited by disk arm

4. C-SCAN



Suppose the requests to be addressed are- 82,170,43,140,24,16,190 and the Read/Write arm is at 50, and it is also given that the disk arm should move "**towards the larger value**".

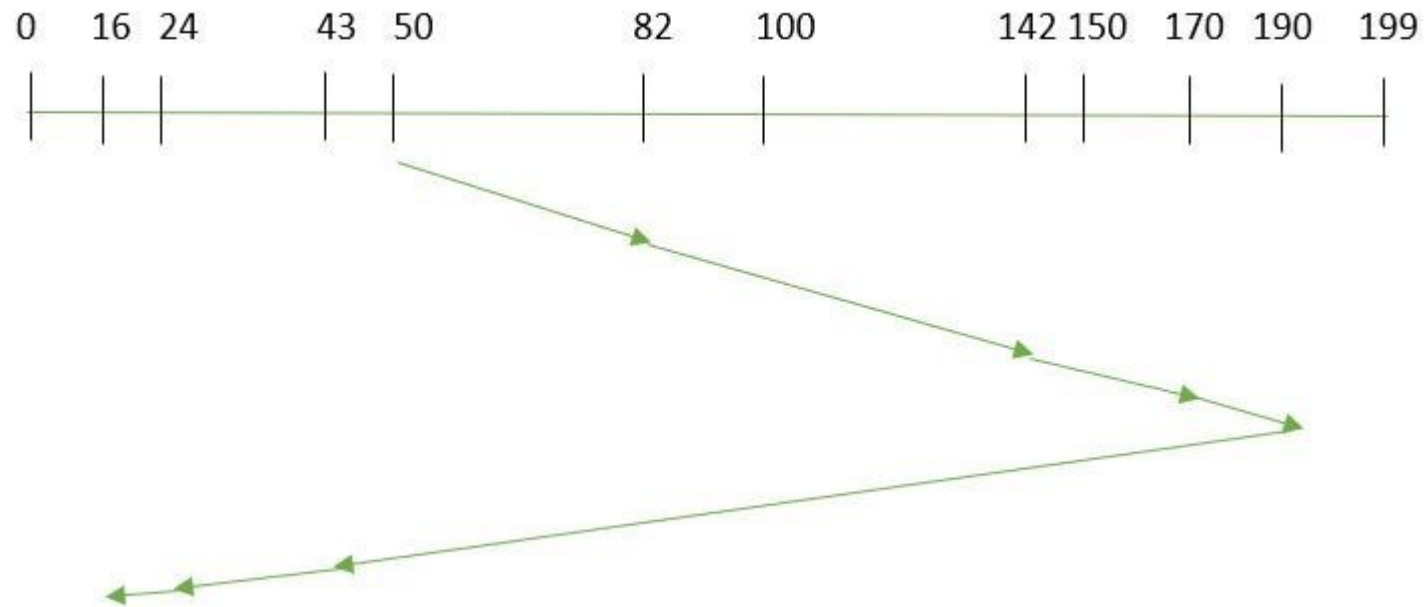
So, the total overhead movement (total distance covered by the disk arm) is calculated as:

$$\begin{aligned} &= (199-50) + (199-0) + (43-0) \\ &= 391 \end{aligned}$$

Advantages of C-SCAN Algorithm:

- Provides more uniform wait time compared to SCAN.

5. LOOK



Suppose the requests to be addressed are- 82,170,43,140,24,16,190 and the Read/Write arm is at 50, and it is also given that the disk arm should move "towards the larger value".

So, the total overhead movement (total distance covered by the disk arm) is calculated as:

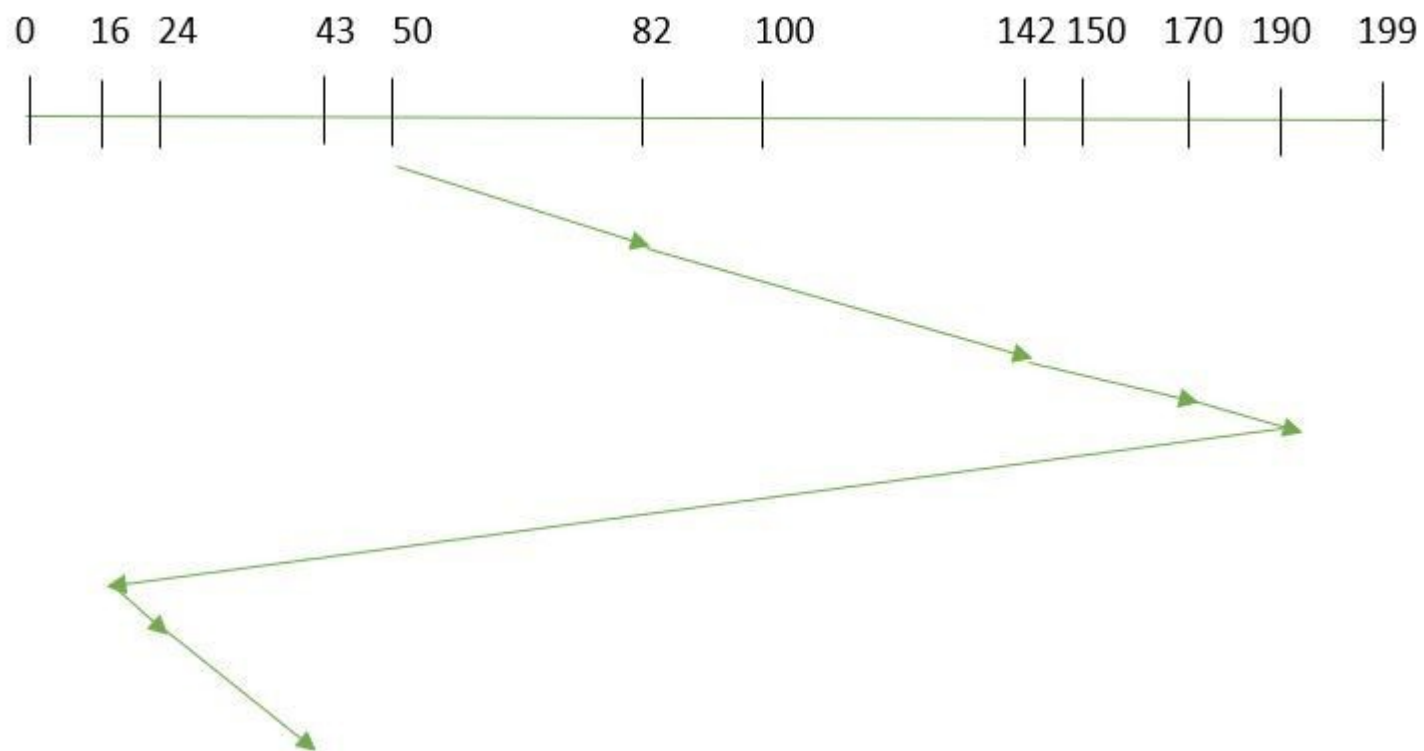
$$= (190-50) + (190-16)$$

$$= 314$$

Advantages

- **Reduced Unnecessary Movement:** The disk arm only goes as far as the last request in each direction, avoiding travel to the disk's physical end (unlike SCAN).
- **Faster Response:** Less head movement leads to quicker service for requests.

6. C-LOOK



So, the total overhead movement (total distance covered by the disk arm) is calculated as

$$= (190-50) + (190-16) + (43-16)$$

$$= 341$$

Advantages:

- **Uniform Wait Time:** Requests are serviced in a circular manner, so waiting times are more predictable and fair.
- **Reduced Head Movement:** The arm only goes as far as the last request in one direction, then jumps back, saving time compared to C-SCAN.